

How do I backup and restore galera clusters in Red Hat OpenStack Services on OpenShift

Issue

As an operator I need to backup the contents of all the galera clusters that are in use in the Openstack control plane.

Resolution

In Red Hat OpenStack Services on OpenShift (RHOSO) 18, several galera clusters can be deployed on the OpenStack control plane, each cluster serving a dedicated set of OpenStack services. For instance, one galera cluster could process traffic for a specific nova cell, while another cluster can process traffic from all the other OpenStack services, such as keystone, cinder, etc.

As an operator, backing up the state of the galera clusters consists of taking a different backup of every galera cluster configured in your OpenStack control plane. You can run the following command to list all the galera clusters deployed in your control plane:

Raw

None

```
oc get galera --no-headers -o custom-columns=":metadata.name"
```

Backing up OpenStack data from a galera cluster

To backup the state of the galera cluster, you select a galera pod and back up all OpenStack databases and tables present in the mysql server running on this pod.

This procedure attempts to select a galera pod that is currently not serving traffic to limit any risks of impacting service during the backup.

The backup file is saved on your workstation in the directory in which you perform this procedure.

Procedure

1. Use the `cd` command to change your current directory to the directory on your workstation where you intend to save the backup.
2. Specify the name of the galera cluster to backup:
3. Raw

None

4.

```
DBNAME=<galera-cluster>
```

5.

Replace `<galera-cluster>` with the name of your galera cluster, for example, `openstack`.

6. Locate the galera pod on this galera cluster that is currently serving traffic:

7. Raw

None

8.

```
SVC_ENDPOINT=$(oc get svc $DBNAME -o json | jq -r  
' .spec.selector["statefulset.kubernetes.io/pod-name"]')
```

9.

List the clustered pods that are currently ready to provide backups:

10. Raw

None

11.

```
ALL_PODS=$(oc exec -q -c galera $SVC_ENDPOINT -- bash -c  
"mysql -nNE -uroot -p\\$DB_ROOT_PASSWORD -e \"show status  
like 'wsrep_incoming_addresses';\" | tail -1 | tr ','  
'\n' | sed 's/\\..*//')
```

12.

Select the target pod that is not currently serving traffic:

13. Raw

None

14.

```
BACKUP_POD=$(echo "$ALL_PODS" | grep -v $SVC_ENDPOINT |  
head -1) BACKUP_POD=${BACKUP_POD:-$DBNAME-galera-0}
```

15.

Retrieve the OpenStack data to backup:

16.Raw

None

17.

```
BACKUP_DBS=$(oc exec -q -c galera $BACKUP_POD -- bash -c  
"mysql -uroot -p\\$DB_ROOT_PASSWORD -sN -e \"select  
distinct table_schema from information_schema.tables  
where engine='innodb' and table_schema != 'mysql';\"")
```

18.

Backup the state of the galera cluster on your workstation:

19.Raw

None

20.

```
BACKUPDATE=$(date -u +%Y-%m-%d_%H-%M-%S') for db in  
$BACKUP_DBS; do echo $db; oc exec -q -c galera  
$BACKUP_POD -- bash -c "mysqldump -uroot  
-p\\$DB_ROOT_PASSWORD --single-transaction --databases  
$db" >> $DBNAME-databases-$BACKUPDATE.sql ;done
```

21.

If the backup is successful, a backup file is saved on your workstation in the current directory. Each backup file preserves the current timestamp, for example `openstack-databases-2025-10-24_10-02-24.sql`.

Restore OpenStack data in a galera cluster

To restore a backup of the OpenStack data to a running galera cluster, you must scale the galera cluster down to 1 pod and disable the database traffic when the data is being restored. When the data is restored you must enable the database traffic and scale the

galera cluster back to 3 pods. Scaling the galera cluster requires updating the existing `OpenStackControlPlane` custom resource (CR).

NOTE: You must perform a full password rotation for every Openstack service declared in the database to restore their OpenStack credentials. See [Restore OpenStack credentials in a galera cluster](#).

Procedure

1. Use the `cd` command to change your current directory to the directory on your workstation in which you have previously saved your galera cluster backups.
2. Specify the name of the galera cluster to restore:
3. Raw

None

4. `DBNAME=<galera-cluster>`

5. Replace `<galera-cluster>` with the name of your galera cluster, for example, `openstack`.
6. Specify the openstack control plane resource that manages the galera cluster:
7. Raw

None

8. `CONTROLPLANE=<control-plane-cr>`

9. Replace `<control-plane-cr>` with the name of your existing `OpenStackControlPlane` CR, for example, `openstack-control-plane`.
10. Specify the name of the galera backup file you want to restore:
11. Raw

None

12. `BACKUP_FILE=openstack-databases-2025-10-24_10-02-24.sql`

13. Scale the galera cluster to 1 pod:

14.Raw

None

15.

```
oc patch oscp $CONTROLPLANE --type merge -p
'{"spec":{"galera":{"templates":{"'$DBNAME'":{"replicas":
1}}}}}' oc wait sts $DBNAME-galera
--for=jsonpath='{.status.replicas}'=1 oc wait sts
$DBNAME-galera
--for=jsonpath='{.status.availableReplicas}'=1
```

16.

Disable the database traffic to this galera cluster during the restore:

17.Raw

None

18.

```
oc patch svc $DBNAME --type=merge -p
'{"spec":{"selector":{"statefulset.kubernetes.io/pod-name
": ""}}}'
```

19.

Optional, if you updated the table schemas or created additional databases since the backup was taken, you may want to drop all the existing Openstack databases prior to restoring the backup:

20.Raw

None

21.

```
oc exec -q -c galera $DBNAME-galera-0 -- bash -c "mysql
-uroot -p\${DB_ROOT_PASSWORD} -sN -e \"select distinct
table_schema from information_schema.tables where
engine='innodb' and table_schema != 'mysql';\" | xargs -r
-n1 mysqladmin -uroot -p\${DB_ROOT_PASSWORD} -f drop"
```

22.

Restore the data:

23.Raw

None

24.

```
oc exec -q -i -c galera $DBNAME-galera-0 -- bash -c  
"mysql -uroot -p\${DB_ROOT_PASSWORD}" < $BACKUP_FILE
```

25.

Verify that the databases are present in the galera pod:

26.Raw

None

27.

```
oc exec -q -it -c galera $DBNAME-galera-0 -- bash -c  
"mysql -uroot -p\${DB_ROOT_PASSWORD} -e \"show  
databases;\""
```

28.

Enable the database traffic to this galera cluster:

29.Raw

None

30.

```
oc patch svc $DBNAME --type=merge -p  
'{"spec":{"selector":{"statefulset.kubernetes.io/pod-name  
": "'$DBNAME-galera-0' "}}}'
```

31.

Scale the galera cluster back to 3 pods:

32.Raw

None

33.

```
oc patch oscp $CONTROLPLANE --type merge -p  
'{"spec":{"galera":{"templates":{"'$DBNAME':"{"replicas":  
3}}}}}' oc wait sts $DBNAME-galera  
--for=jsonpath='{.status.availableReplicas}'=3
```

Restore OpenStack credentials in a galera cluster

You must perform a full password rotation for every Openstack service declared in the database to restore their OpenStack credentials. In RHOSO 18, there is one MariaDBAccount CR for the Openstack user defined in a galera cluster.

This is the high-level password rotation workflow for a galera cluster:

1. Create a new MariaDBAccount CR for each Openstack service, to automatically create new credentials in the database, without deleting the existing ones.
2. For each service defined in the control plane, reference the name of the new MariaDBAccount in the service's CR. This automatically triggers a rolling restart of the Openstack service to prevent service disruption.
3. When all services have been restarted, delete the old MariaDBAccount CRs.